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Münster

Can Tobacco Control Policies Prevent Tooth Loss? A Difference-in-Differences Study

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Sebastian-Edgar Baumeister

Benjamin W. Chaffee · Birte Holtfreter · Stefan Listl

- Smoking is the dominant modifiable cause of periodontitis and tooth loss
- Cigarette taxes and prices reduce smoking; the evidence for smoke-free laws is weaker
- Tobacco policy is rarely evaluated against oral-health outcomes

Primary aim (substantive)

Estimate the effect of U.S. state tobacco-control policies on smoking and on tooth loss

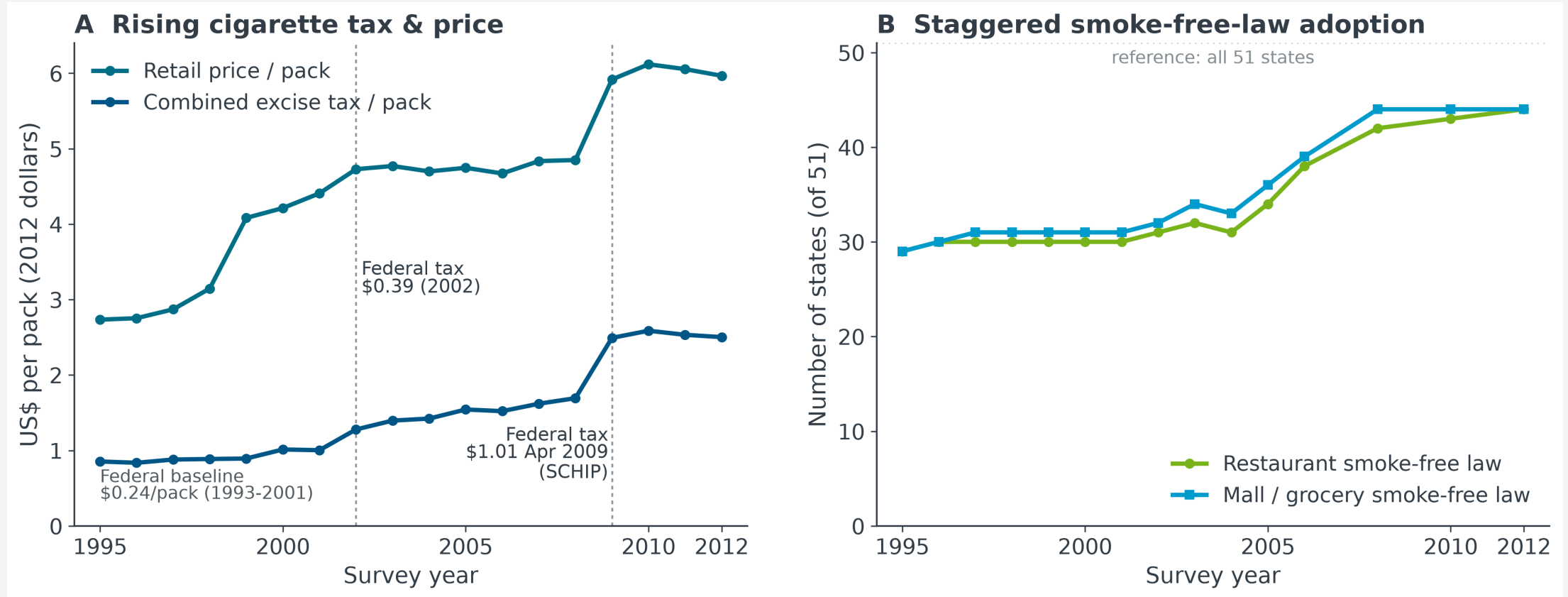
Secondary aim (methodological)

Benchmark heterogeneity-robust DiD estimators against two-way fixed effects in a design combining staggered binary and continuous treatments

	Operationalisation
Sample	Behavioral Risk Factor Surveillance System (BRFSS), 1995–2012; 15 cycles, 51 states; analytic N = 2,622,010 (smoking) and 2,638,355 (tooth loss)
Treatments	Cigarette tax and price (\$/pack, continuous); restaurant and mall/grocery smoke-free laws (staggered)
Outcomes	Complete edentulism; loss of ≥ 6 teeth
Control outcomes	Positive: current smoking, quitting; negative: seat-belt use, smoke-detector

U.S. state tobacco-control policies, 1995–2012

A rising cigarette tax and price (left) and smoke-free-law adoption (right) — the variation the difference-in-differences design exploits



Two-way fixed effects: the model, its estimand, and where it fails here

$$Y_{ist} = \beta D_{st} + X'_{ist}\gamma + \alpha_s + \lambda_t + \varepsilon_{ist}$$

i individual, s state, t year. D = policy (dose in \$/pack, or ban vs none); X = covariates; α = state fixed effects; λ = year fixed effects. $\beta \times 100$ is in percentage points — per +\$1.00 of dose, or for ban versus no ban

Target estimand: the average treatment effect on the treated, identified under parallel trends in the mean untreated potential outcome

Staggered binary treatment — the smoke-free laws

- β is a weighted average of every observed 2×2 DiD; the weights are non-negative (Goodman-Bacon 2021), but already-treated states serve as controls — contaminated comparisons
- Rewritten over group-time effects the weights need not be non-negative (de Chaisemartin–D'Haultfœuille 2020): the sign of the estimand is not guaranteed

Continuous dose — cigarette tax and price

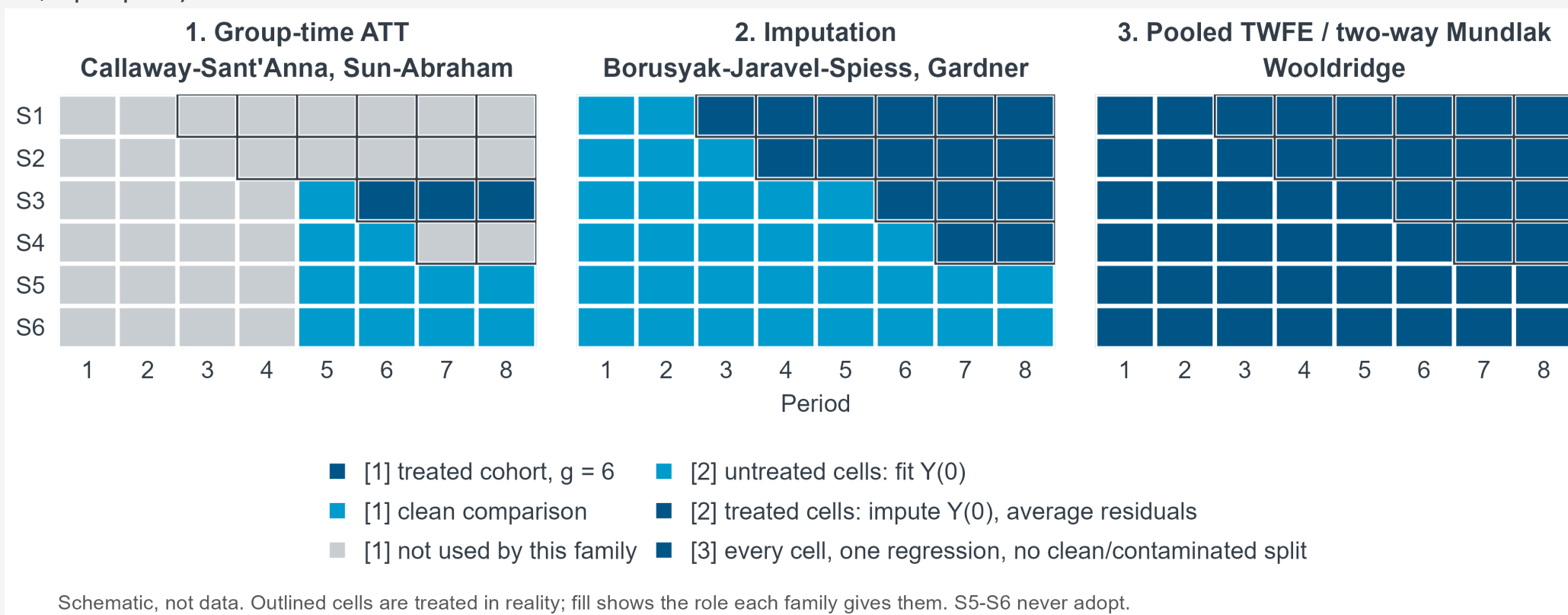
- Dose in constant 2012 dollars: every state's real dose moves every year — no stayers
- No untreated cells after 1996: year effects for 1997–2012 unidentified, so the secular decline is absorbed into the dose coefficient

$$ATT(d_j | d_j) - ATT(d_{j-1} | d_{j-1}) = \underbrace{ACRT(d_j | d_j) \cdot \Delta d}_{\text{causal}} + \underbrace{ATT(d_{j-1} | d_j) - ATT(d_{j-1} | d_{j-1})}_{\text{selection on dose}}$$

Comparing dose levels returns the causal response plus selection on dose (Callaway et al. 2024)

Heterogeneity-robust estimators: which cells each family compares

The three families differ in one thing that matters: what they are willing to use as a comparison. Continuous dose handled as high-dose cohorts (tax $\geq \$3$, price $\geq \$6$ per pack)



No clean comparison for the continuous dose: the baseline-deviation specification implies -3.1 pp per $\$1.00$ on smoking — roughly seven times the adequately identified estimate

Results: cigarette tax, smoking, and tooth loss

Smoking — percentage points (95% CI) per +\$1.00/pack

Outcome	Tax, current year	Tax, 10-year average
Current smoking	−0.35 (−0.56, −0.13)	−0.47 (−0.95, 0.01)
Quitting	0.51 (0.19, 0.82)	0.07 (−0.53, 0.67)

Tooth loss — percentage points (95% CI) per +\$1.00/pack

Outcome	Tax, current year	Tax, 10-year average
Complete edentulism	0.08 (−0.14, 0.31)	0.39 (−0.12, 0.89)
Loss of ≥6 teeth	−0.19 (−0.53, 0.15)	−0.44 (−1.17, 0.30)

Covariate-adjusted two-way (state and year) fixed-effects estimates, linear probability model, unweighted individual-level data; standard errors clustered by state. Retail price shows similar estimates.

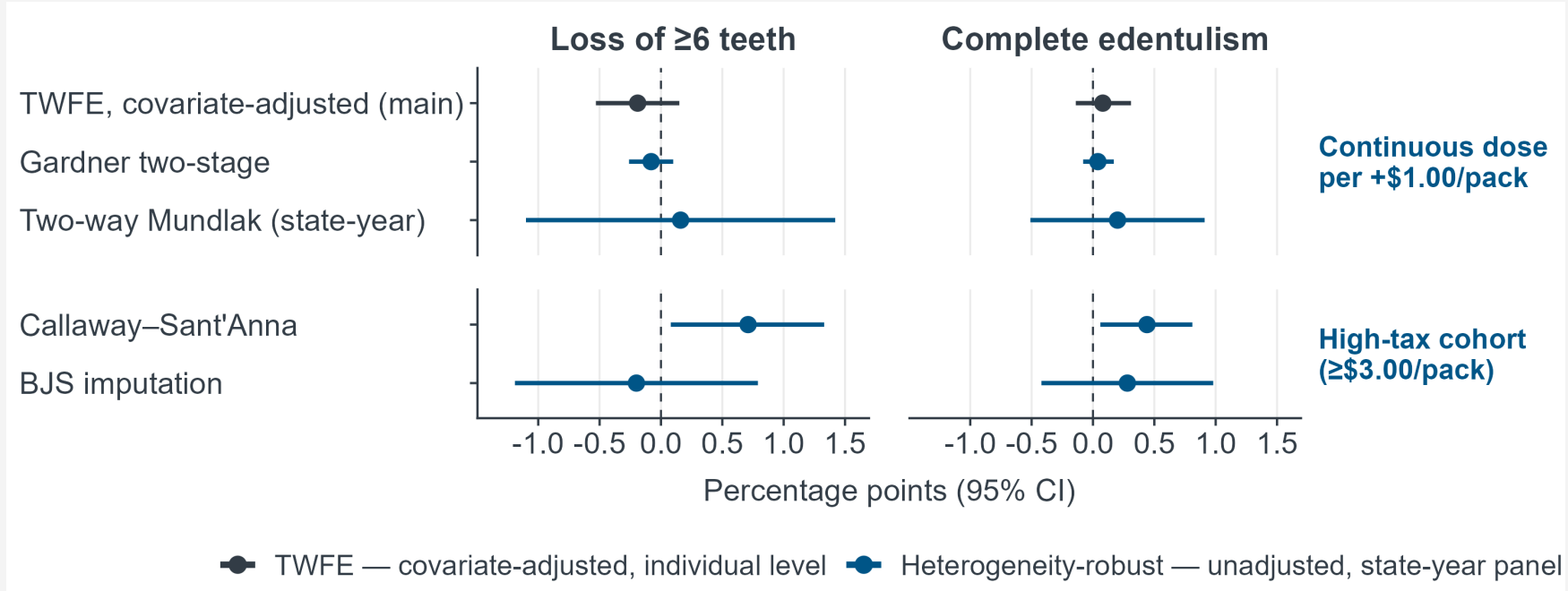
Smoke-free laws: estimates that fail their own falsification test

Percentage points (95% CI), ban vs no ban

Ban	Current smoking	Complete edentulism	Loss of ≥ 6 teeth
Restaurant	0.11 (−0.34, 0.55)	0.45 (0.03, 0.88)	0.51 (−0.02, 1.05)
Mall / grocery	0.30 (−0.21, 0.81)	0.24 (−0.19, 0.68)	0.52 (−0.15, 1.18)

- No first stage: neither ban lowered smoking, and several tooth-loss estimates are positive
- The mall/grocery ban fails its negative control: smoke-detector test −1.17 pp (−2.27, −0.08)
- The tax passes both negative controls (seat-belt 0.77; smoke detector 1.75, both null)

Estimator comparison: where the methods agree and diverge



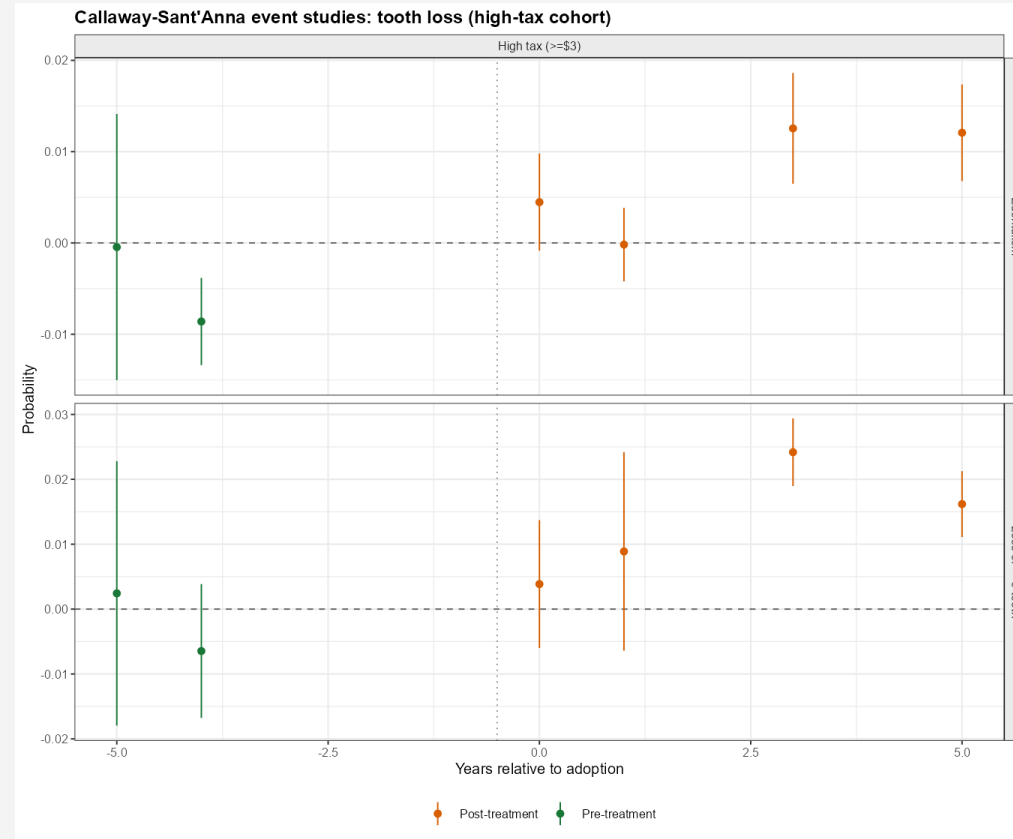
Why the estimates differ

- Callaway-Sant'Anna fails its own positive control: on the same high-tax cohort it gives +0.93 pp for smoking (−1.76, 3.63)
- Different estimands: continuous per +\$1.00 dose versus binary $\geq \$3.00$ -cohort ATT
- Callaway-Sant'Anna rests on small, late, biennially observed cohorts; imputation, more stable, stays null

Sun-Abraham omitted here: for the high-dose tooth-loss cohorts — small, late and biennially observed — its variance is undefined (the group-time estimator does not converge there either); it is estimable for the bans

Callaway–Sant'Anna event study (Roth 2026)

Universal base period (Roth 2026). Pre-treatment coefficients are not negligible — -0.6 to -0.9 pp, larger than any effect of interest — and post-treatment coefficients turn significantly positive. Read together: a differential trend, not a protective policy



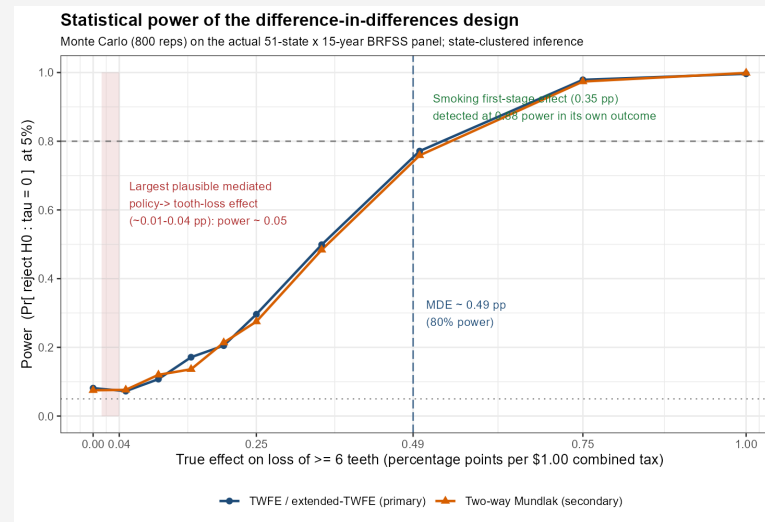
The gaps are structural, not selective: no $t-1$ (universal base period); no $t-3$ or $t-2$ (the oral-health module ran annually to 2006, then every second year); and the high-dose cohorts adopt late

$$E[\Delta Y(0) \mid D = 1] = E[\Delta Y(0) \mid D = 0]$$

- A pre-trend of the size just shown is exactly what this assumption forbids
- Untestable: pre-trend tests have low power, and conditioning on passing one distorts inference (Roth 2022) — judge magnitude
- Identification also needs a stable-composition condition; both are conditions on population means, not individual trajectories
- The causal graph locates the assumption but does not establish it (Renson et al. 2026); the Δ -SWIG is Knaus & Pfleiderer's (2026) device — backup slide
- Rambachan–Roth relative magnitudes ($\bar{M} = 2$): the ≥ 6 -teeth CI widens from $[-0.20, 1.50]$ to $[-9.0, 11.1]$ pp

Is the null informative? Minimum detectable effect and plausibility

- CI excludes protective effects beyond ≈ 0.5 pp contemporaneous, ≈ 1.2 pp cumulative — a precise null, not an inconclusive one
- Minimum detectable effect 0.49 pp, against a plausible mediated effect of 0.01–0.04 pp — 10–40 $\times$ below it
- The ceiling is between-state variation and 51 clusters, not the 3.8 m respondents (Bertrand et al. 2004)



An equivalence statement, not evidence of absence (Rainey 2014; Lakens et al. 2018)

1. Cigarette taxes and prices reduce smoking; no detectable effect on tooth loss over 1995–2012, though sustained exposure and severe tooth loss warrant evaluation over longer horizons
2. The null is a limit of the design: the plausible mediated effect lies far below the minimum detectable effect
3. For tooth loss the robust estimators diverge: imputation, two-stage and extended TWFE null, Callaway–Sant'Anna positive on sparse late high-dose cohorts
4. Continuous-dose DiD is the binding constraint: no stayers, and dose contrasts carry selection on dose



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Thank you

Paper in print in the *Journal of Dental Research*

Slides and analysis code:

github.com/BaumeisterS/DGEPI2026_tobacco_policy_tooth_loss

DiD estimators: Goodman-Bacon A. 2021. Difference-in-differences with variation in treatment timing. *J Econom.* 225(2):254–277. de Chaisemartin C, D'Haultfœuille X. 2020. Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *Am Econ Rev.* 110(9):2964–2996. Callaway B, Sant'Anna PHC. 2021. Difference-in-Differences with multiple time periods. *J Econom.* 225(2):200–230. Sun L, Abraham S. 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J Econom.* 225(2):175–199. Borusyak K, Jaravel X, Spiess J. 2024. Revisiting Event-Study Designs: Robust and Efficient Estimation. *Rev Econ Stud.* 91(6):3253–3285. Gardner J. 2022. Two-stage differences in differences. *arXiv:2207.05943*. Liu L, Wang Y, Xu Y. 2024. A Practical Guide to Counterfactual Estimators for Causal Inference with Time-Series Cross-Sectional Data. *Am J Polit Sci.* 68(1):160–176. Wooldridge JM. 2025. Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators. *Empir Econ.* 69(5):2545–2587. de Chaisemartin C, D'Haultfœuille X, Vazquez-Bare G. 2024. Difference-in-Difference Estimators with Continuous Treatments and No Stayers. *AEA Pap Proc.* 114:610–613. Callaway B, Goodman-Bacon A, Sant'Anna PHC. 2024. Difference-in-differences with a Continuous Treatment. NBER Working Paper 32117.

Identification, sensitivity and reviews: Roth J. 2022. Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends. *Am Econ Rev Insights.* 4(3):305–322. Roth J. 2026. Interpreting event-studies from recent difference-in-differences methods. *Jpn Econ Rev.* 77(2):275–288. Rambachan A, Roth J. 2023. A More Credible Approach to Parallel Trends. *Rev Econ Stud.* 90(5):2555–2591. Roth J, Sant'Anna PHC, Bilinski A, Poe J. 2023. What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *J Econom.* 235(2):2218–2244. Baker A, Callaway B, Cunningham S, Goodman-Bacon A, Sant'Anna PHC. 2026. Difference-in-Differences Designs: A Practitioner's Guide. *J Econ Lit.* 64(2):498–557. Knaus MC, Pfliederer H. 2026. Causal Graphs for Conditional Parallel Trends. *arXiv:2604.12818*. Renson A, Dukes O, Shahn Z. 2026. Using Causal Diagrams to Assess Parallel Trends in Difference-in-Differences Studies. *Stat Med.* 45(6–7):e70459.

Power, precision and equivalence: Bertrand M, Duflo E, Mullainathan S. 2004. How Much Should We Trust Differences-In-Differences Estimates? *Q J Econ.* 119(1):249–275. Rainey C. 2014. Arguing for a Negligible Effect. *Am J Polit Sci.* 58(4):1083–1091. Lakens D, Scheel AM, Isager PM. 2018. Equivalence Testing for Psychological Research: A Tutorial. *Adv Methods Pract Psychol Sci.* 1(2):259–269.

Substantive: GBD 2019 Tobacco Collaborators. 2021. Spatial, temporal, and demographic patterns in prevalence of smoking tobacco use and attributable disease burden in 204 countries, 1990–2019. *Lancet.* 397(10292):2337–2360. Chaloupka FJ, Powell LM, Warner KE. 2019. The Use of Excise Taxes to Reduce Tobacco, Alcohol, and Sugary Beverage Consumption. *Annu Rev Public Health.* 40:187–201. Baumeister S-E, Listl S, Holtfreter B, Nascimento GG, Leite FRM. 2025. Causal Effect of Smoking and Cessation on Tooth Loss. *J Clin Periodontol.* 52(4):539–546. Tonetti MS, Greenwell H, Kornman KS. 2018. Staging and grading of periodontitis. *J Clin Periodontol.* 45 Suppl 20:S149–S161.

Backup: what the estimand averages over

A hostile question this pre-empts: "isn't your effect diluted, since most people never change behaviour?"

- The target is the population average treatment effect on the treated — for the continuous dose, the corresponding average causal derivative
- The policy is assigned at the state–year level, and its evaluation averages over the entire population it governs, the large majority of whom do not change behaviour
- So dilution is not a flaw in the estimate; it is the quantity a policymaker needs
- Identification: parallel trends in the mean untreated potential outcome, plus a stable-composition condition — conditions on population means, not individual trajectories

Backup: the three estimator families side by side

Family	Comparison group	What it needs	What it gave here
Group-time ATT Callaway–Sant'Anna 2021; Sun–Abraham 2021	Not-yet-treated (CS); the last-treated cohort where no never-treated exist (SA)	No anticipation + parallel trends; the two rest on non-nested assumptions	Positive and significant on the high-tax cohort — but fails its own positive control
Imputation Borusyak–Jaravel–Spiess 2024; Gardner 2022	All untreated cells, which fit the $Y(0)$ model	Untreated cells in every period, or the year effects are unidentified	Null throughout; the most stable family on this panel
Pooled TWFE / two-way Mundlak Wooldridge 2025	None — every cell enters one regression	Homogeneous effects, or a sufficiently flexible model	Null; reported as a specification check, not a remedy

$$ATT(g, t) = E[Y_t(g) - Y_t(0) \mid G_g = 1]$$

The group-time estimand. It is *identified* — not defined — by the difference-in-differences contrast against a clean comparison cohort, under no anticipation and parallel trends. Wooldridge (2025) shows the two-way Mundlak regression reproduces TWFE for *unweighted* pooled OLS on complete cases; the row reported here was run on collapsed, weighted state-year means, so it is a specification check rather than a literal restatement of TWFE

Backup: what TWFE estimates under staggered adoption

The two decompositions behind slide 5 — the weights are non-negative in the first and need not be in the second

$$\hat{\beta}^{\text{TWFE}} = \sum_k s_k \widehat{DD}_k, \quad s_k \geq 0, \quad \sum_k s_k = 1$$

Goodman-Bacon (2021): a variance-weighted average of all observed 2×2 DiDs. The weights are variance shares, so they cannot be negative; the bias enters because some 2×2 comparisons use already-treated units as controls

$$\beta^{\text{TWFE}} = \sum_{g,t} w_{g,t} \text{ATT}(g,t), \quad \exists w_{g,t} < 0$$

de Chaisemartin–D'Haultfœuille (2020): rewritten over group-time effects, the weights need not be non-negative — so β can carry the wrong sign even when every underlying $\text{ATT}(g,t)$ has the same sign

Backup: inference with 51 clusters

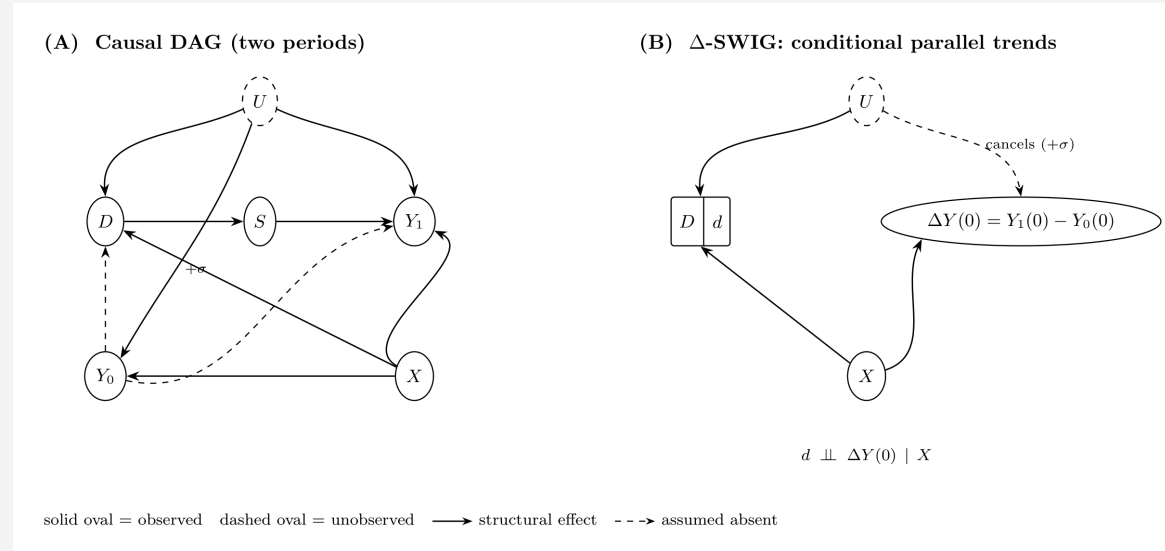
- Standard errors clustered by state throughout (51 clusters)
- Monte Carlo on the actual 51-state $\times$ 15-year panel (800 replications): power ≈ 0.50 , 0.77 and 0.98 at true effects of 0.35, 0.50 and 0.75 pp
- The simulation shows a mild finite-cluster over-rejection (size ≈ 0.08) — conservative for a no-effect conclusion
- A CR2 bias-reduced, Satterthwaite-corrected small-cluster variance leaves the intervals essentially unchanged, confirming the 51-cluster inference is not anti-conservative
- The equivalence bound is read from the interval itself, not from the simulated size

Backup: survey design, weighting and the survey years

- Models are fitted to unweighted individual-level data; BRFSS survey weights were not applied
- The estimand is the state–year policy effect, identified from within-state variation net of state and year fixed effects
- 15 cycles: 1995–2006 annually, then 2008, 2010 and 2012
- 2007, 2009 and 2011 are absent because the oral-health module was not administered in those cycles — not a selection decision
- Analytic N $\approx$ 2.6 m (2,622,010 smoking; 2,638,355 tooth loss) of 3,840,519 pooled respondents

Backup: the causal graph and the Δ -SWIG

U = time-invariant unmeasured state factors (oral-health infrastructure, dental workforce, fluoridation history). The Δ -SWIG adds an explicit difference node, so $d \perp \Delta Y(0) \mid X$ read off the graph IS conditional parallel trends



Not formally addressed in the paper — the argument, not a published result

- Interference is plausible: residents of a high-tax state can buy across a state line, so a neighbour's consumption depends on this state's tax
- Direction: cross-border purchasing weakens the realised dose in treated states, attenuating the first stage toward the null
- The first stage is nonetheless detected (-0.35 pp per \$1.00), so attenuation does not overturn it
- For tooth loss the design is already far short of the minimum detectable effect, so interference is not what produces the null